

README

scripts/caf/ — Fork-ALL cli_agents + auto-merge mechanism (SP3)

Reusable, fully-parameterized bash scripts that fork every third-party cli_agents/* submodule under an org with a prefix, keep each fork current by merging upstream main/master, and recursively classify nested deps.

Nothing is hardcoded — org / prefix / src-dir / providers / branch / depth are all flags or CAF_* env vars (see caf_lib.sh).

Files

File	Purpose	External mutation?
caf_lib.sh	shared helpers (URL normalize, own-org test, fork-name, branch-detect, caf_safe_push no-force guard, arg parsing)	none (sourced)
fork_third_party_submodule.sh	gh+glab fork-or-create under --org/--prefix, wire origin=fork + upstream=original, record --map-file	YES — OPERATOR-GATED (G-1)
update_fork_from_upstream.sh	fetch + merge upstream main/master INTO each fork, fast-forward/merge only, push fan-out	YES (push) — OPERATOR-GATED (G-1-push)
resolve_recursive_fork_deps.sh	recursive nested-dep classifier: third-party → FORK:, own-	report-only by default;

File	Purpose	External mutation?
	org → PULL_TO_ROOT (CONST-051)	delegated fork is G-1
challenge_caf.sh	anti-bluff Challenge — proves all of the above against throwaway LOCAL git repos (no real remote) + paired §1.1 mutation	none (mktemp scratch only)

Binding constraints (cited, never weakened)

- §11.4.113 — absolute no-force-push. Every push goes through `caf_safe_push`, which **refuses** `--force` / `--force-with-lease` / `--mirror` / `+ref` overwrites. Merge is always upstream → fork on top of the fork tip, so every push is a fast-forward — force is never needed.
- §11.4.28(B) / **CONST-051** — forks stay decoupled / project-not-aware. The ONLY edits to a fork are non-source: remote config, `.gitignore`, governance pointers, nested-`.gitmodules` SSH rewrites. Never a blanket `git add -A`.
- **Rule 1 (No CI/CD)** — auto-merge is host-local (launchd/cron) only; no `.github/workflows`, `.gitlab-ci.yml`, etc.
- **Rule 3 (SSH only)** — every fork URL and rewritten nested URL is `git@...`
- §11.4.101 — every irreversible external mutation (fork creation, push) is OPERATOR-GATED behind **G-1**; the agent makes only reversible local decisions autonomously. The **default is --dry-run**.
- **CONST-033** — never touches host power state.

Usage

Plan only (DEFAULT — mutates nothing, no remote touched):

```
bash scripts/caf/fork_third_party_submodule.sh --dry-run
```

OPERATOR-GATED (G-1) — actually create the forks (requires gh; glab for gitlab):

```
bash scripts/caf/fork_third_party_submodule.sh --execute \
  --org vasic-digital --prefix caf- --src-dir cli_agents --
  providers github,gitlab
```

Auto-merge unit of work (G-1-push for the push step):

```
bash scripts/caf/update_fork_from_upstream.sh --map-file docs/
  caf/map.tsv --execute
```

```
# Recursive nested-dep classification (report-only):
bash scripts/caf/resolve_recursive_fork_deps.sh --src-dir
      cli_agents --depth 3

# Anti-bluff Challenge (local-only, no remote) + paired mutation:
bash scripts/caf/challenge_caf.sh          # → 28 PASS, 0 FAIL,
      exit 0
bash scripts/caf/challenge_caf.sh --mutate  # → an assertion
      FAILs (guard is real), exit 1
```

Flag / env surface (all defaults overridable)

```
--org / CAF_ORG                default vasic-digital
--prefix / CAF_PREFIX           default caf-
--src-dir / CAF_SRC_DIR         default cli_agents
--gitmodules / CAF_GITMODULES   default <repo>/..gitmodules
--providers / CAF_PROVIDERS     default github,gitlab
--gitlab-group / CAF_GITLAB_GROUP default vasic-digital
--branch / CAF_BRANCH           default "" → auto-detect remote HEAD (main
--only / CAF_ONLY               restrict to a subset (resumable retry)
--exclude / CAF_EXCLUDE         default claude-code-source
--depth / CAF_DEPTH             default 3 (recursion bound)
--workdir / CAF_WORKDIR         default mktemp -d
--map-file / CAF_MAP_FILE       default docs/caf/map.tsv
--strategy / CAF_STRATEGY       merge (default) | ff-only
--visibility / CAF_VISIBILITY   private (default) | public
--dry-run (default) | --execute / --no-dry-run
--push (default) | --no-push
```

Graceful degradation

- gh/ghlab absent → the real-remote fork path SKIPS with a logged reason and exits 2; --dry-run always works.
- update_fork_from_upstream.sh and resolve_recursive_fork_deps.sh need only git.
- challenge_caf.sh uses **only local git** — it requires no gh/ghlab and touches no real remote.

Auto-merge schedule (host-local, NO CI per Rule 1)

update_fork_from_upstream.sh is the scheduled unit of work. NOT committed as a project pipeline.

macOS (launchd) — ~/Library/LaunchAgents/digital.vasic.caf-update.plist, weekly, runs detached:

```
<key>ProgramArguments</key>
<array>
  <string>/opt/homebrew/bin/bash</string>
  <string>/Volumes/T7/Projects/HelixCode/scripts/caf/
    update_fork_from_upstream.sh</string>
  <string>--map-file</string><string>/Volumes/T7/Projects/
    HelixCode/docs/caf/map.tsv</string>
  <string>--execute</string>
</array>
<key>StartCalendarInterval</key><dict><key>Weekday</
  key><integer>1</integer><key>Hour</key><integer>4</
  integer></dict>
```

Linux (cron) — @weekly:

```
@weekly /usr/bin/env bash -lc 'nohup bash $HOME/Projects/HelixCode/scripts
```

Evidence / ledgers

- Run ledger: appended to docs/caf/Status.md (append-only).
- Conflicts: parked to docs/caf/conflicts/<fork>_<ts>.md — **never** auto-resolved (no -s ours / -X theirs).
- The Challenge writes only to a mktemp -d scratch tree and self-cleans.

Exit codes

Script	0	1	2
fork_third_party_submodule.sh	all handled / dry-run planned	partial (some FAIL logged)	env (gh/ glab/git missing)
update_fork_from_upstream.sh	all updated, no conflict	partial: a CONFLICT parked / PUSHFAIL	env (git / map-file missing)
resolve_recursive_fork_deps.sh	no CONST-051 violation	own-org nested chain flagged	env (git / scan-root missing)
challenge_caf.sh		≥1 FAIL (incl. expected	—

Script	0	1	2
	all assertions PASS	under - - mutate)	